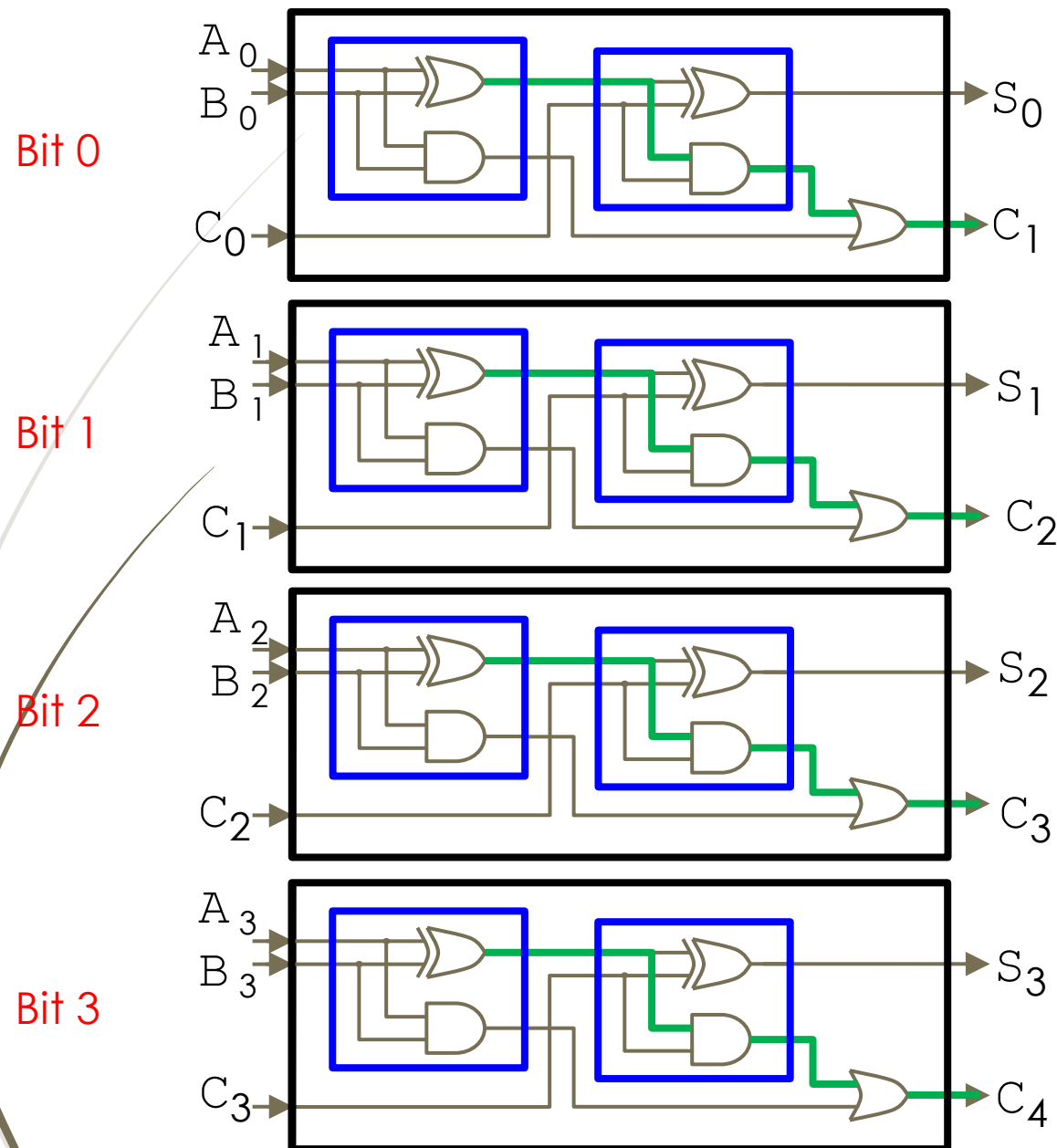
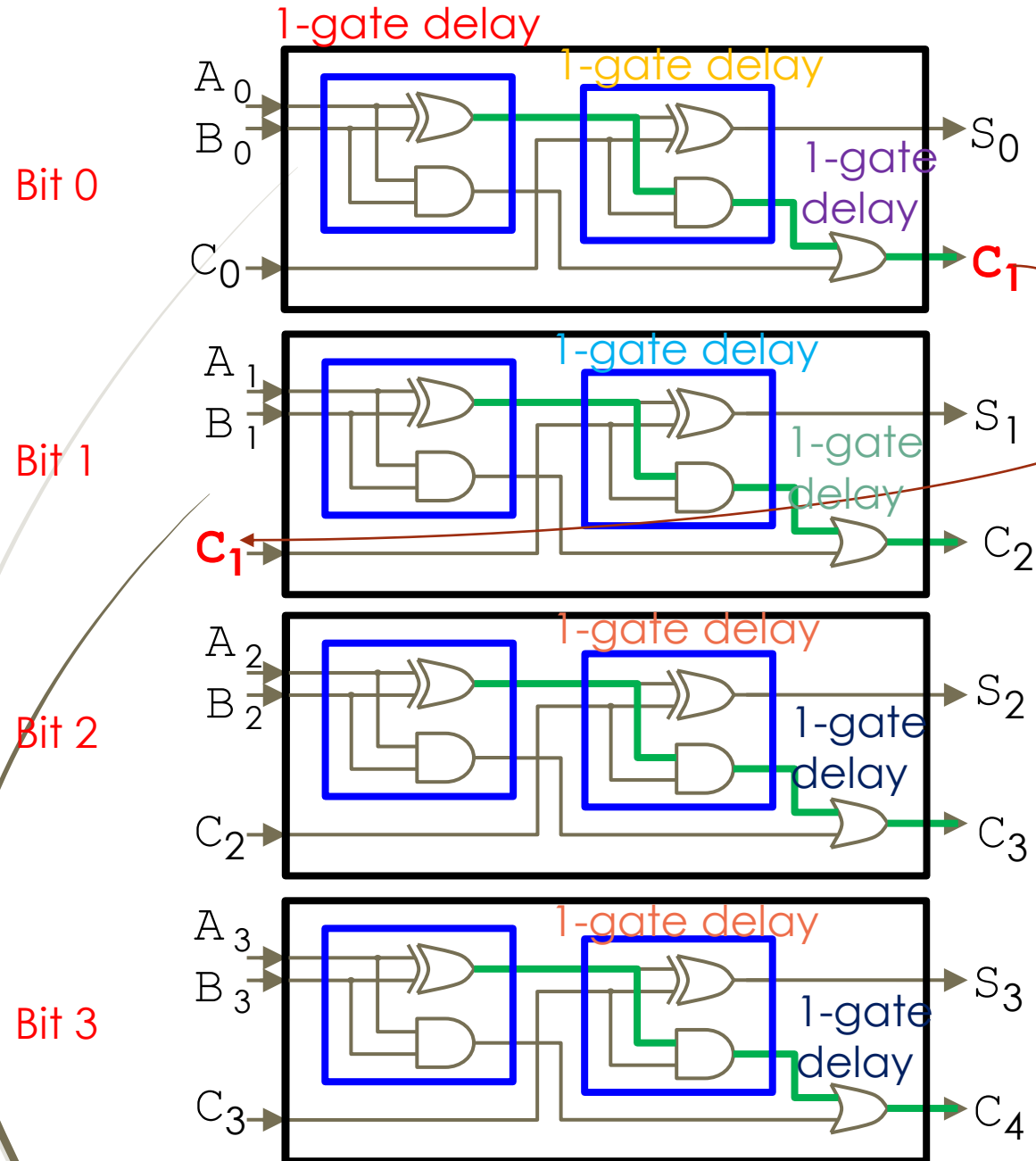


# Challenge

- Why is the propagation delay of this Full Adder 4 (also known as a 4-bit (w-bit) ripple carry adder) 9 gates ( $2w + 1$  gates) where  $w$  is the number of bits (as opposed to 12 gates  $\rightarrow 4 \times 3$  gate p.d.)?





# Solution

**Answer:**

$1 + 4 * 2\text{-gate delay} = 9\text{-gate delay}$   
 where  $w = 4$

**Explanation:**

Imagine you are adding 1000 (8)  

$$\begin{array}{r} 1000 \\ + 1001 \\ \hline 10001 \end{array}$$

So,  $A_3 = 1$ ,  $A_2 = 0$ ,  $A_1 = 0$ ,  $A_0 = 0$ ,  
 $B_3 = 1$ ,  $B_2 = 0$ ,  $B_1 = 0$ ,  $B_0 = 1$ ,  
 $S_3 = 0$ ,  $S_2 = 0$ ,  $S_1 = 0$ ,  $S_0 = 1$  and  
 $C_4 = 1$ ,  $C_3 = 0$ ,  $C_2 = 0$ ,  $C_1 = 0$ ,  $C_0 = 0$ .

This signifies that the 4 bits of A and B are coming into the first half-adder at the same time and producing the output in 1-gate delay for all 4 full-adders at the same time. This is the "1 +".

Looking at the **Bit 0 full-adder**, this output (from first half-adder) becomes an input, along with  $C_0$ , to the second half-adder, which produces  $S_0$  in 1-gate delay. One of its output becomes the input to the OR gate and produces  $C_1$  in 1-gate delay. So, it takes 1-gate delay + 1-gate delay + 1-gate delay to produces  $C_1$ .

In the meantime, the second half-adder of **Bit 1 full-adder** is waiting for  $C_1$  in order to produce  $S_1$  and  $C_2$ . So, it takes 1-gate delay + 1-gate delay + 1-gate delay + 1-gate delay + 1-gate delay to produces  $C_2$ .

In the meantime, the second half-adder of **Bit 2 full-adder** is waiting for  $C_2$  in order to produce  $S_2$  and  $C_3$ . So, it takes 1-gate delay + 1-gate delay + 1-gate delay + 1-gate delay + 1-gate delay + 1-gate delay to produces  $C_3$ .

Finally, the second half-adder of **Bit 3 full-adder** is waiting for  $C_3$  in order to produce  $S_3$  and  $C_4$  and this will also take 2-gate delay.

To conclude, there is the initial 1-gate delay associated with the first half-adder which occurs at the same time for each full-adder, then, for each full-adder, there is 2-gate delay, serially (one after the other). Since we have  $w = 4$  full-adders, the total delay will be  $1 + 4 * 2\text{-gate delay} = 9\text{-gate delay}$ .

# Solution

Bit 0

Bit 1

Bit 2

Bit 3

